Problem set – 11 PAL, PLA

1. Design a BCD-to-XS-3 code converter using a (a) PROM, (b) PLA and (c) PAL

Solution: The conversion table for a BCD to Excess-3 code converter is shown below. Let $B_3\ B_2\ B_1\ B_0$ be the BCD word and $E_3\ E_2\ E_1\ E_0$ be the Excess-3 word.

Conversion Table

BCD				Excess-3			
B_3	B_2	B_1	B_0	E_3	E_2	E_1	E_0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

Since minterms 10, 11, 12, 13, 14, 15 are don't cares in BCD, from the truth table we get the expressions for XS-3 word as

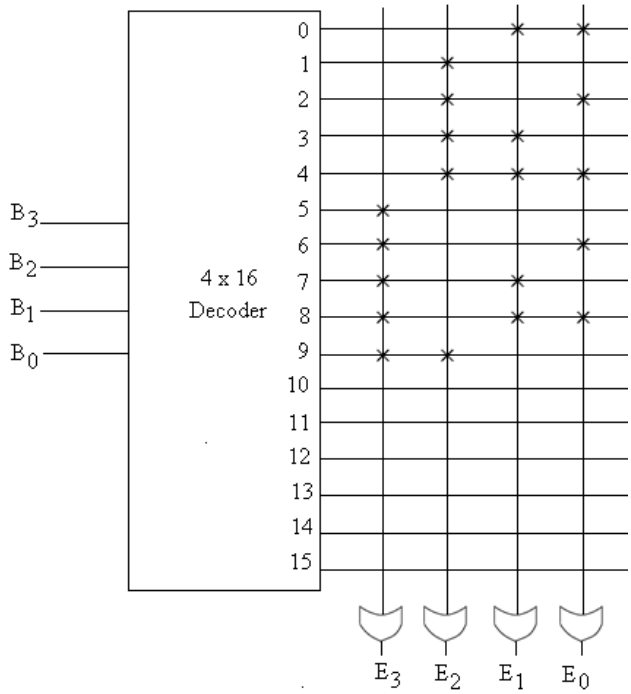
$$E_3 = \sum m\,(5,6,7,8,9) + d(10,11,12,13,14,15)$$

$$E_2 = \sum m\,(1,2,3,4,9) + d(10,11,12,13,14,15)$$

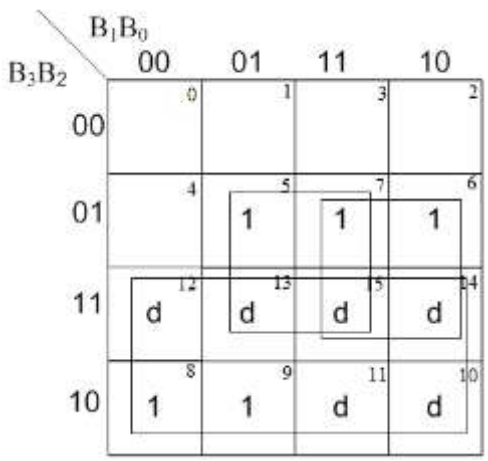
$$E_1 = \sum m\,(0,3,4,7,8) + d(10,11,12,13,14,15)$$

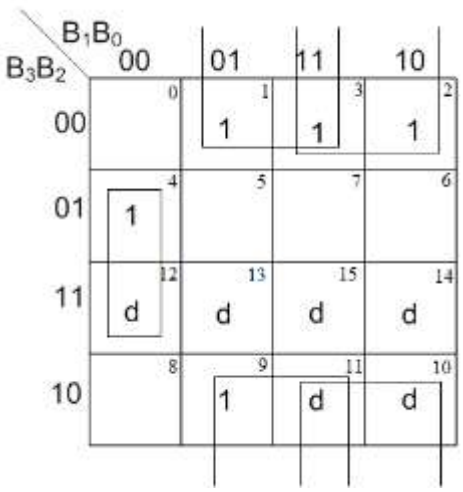
$$E_0 = \sum m\,(0,2,4,6,8) + d(10,11,12,13,14,15)$$

For realization by EPROM, there is no need to minimize the expressions. Since the input is a 4 variable one, we require a 4×16 decoder. Since the number of outputs is 4, we require a 16×4 PROM. The realization of the BCD to Excess-3 code converter using a PROM is shown in figure below.



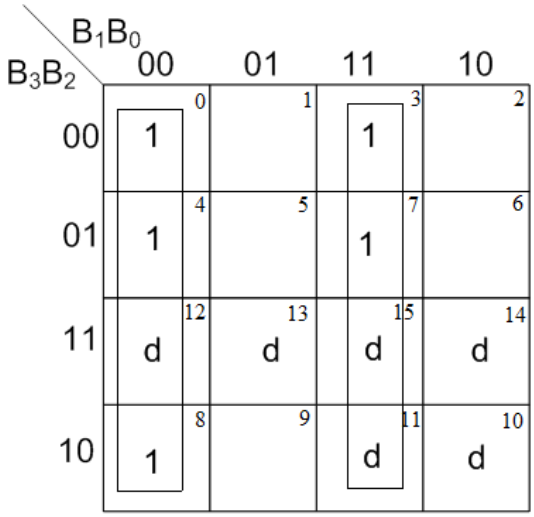
b) Using the expressions for E_3, E_2, E_1 and E_0 given in (a) the minimal expressions for E_3, E_2, E_1 and E_0 are obtained from the K-maps as shown in figure below.

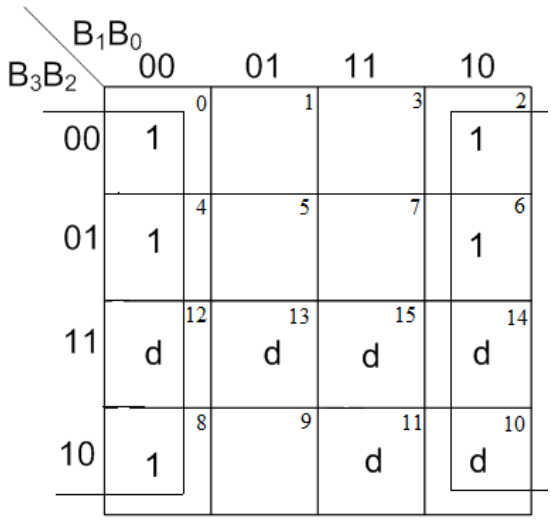




$E_3 = B_3 + B_2B_0 + B_2B_1$

$E_2 = B_2\overline{B_1}\,\overline{B_0} + \overline{B_2}\,B_0 + \overline{B_2}\,B_1$



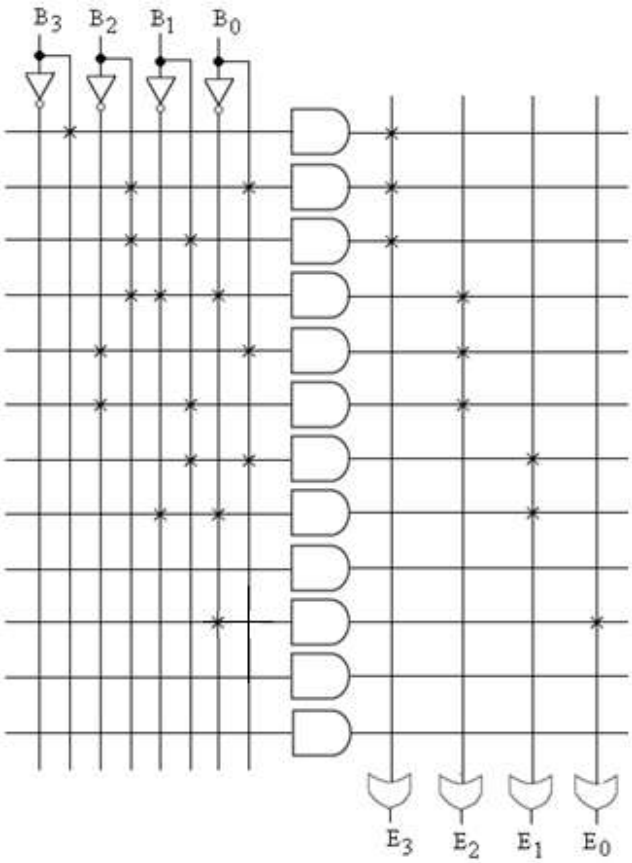


$E_1 = \overline{B_1}\ \overline{B_0} + B_1B_0$ $\qquad$ $E_0 = \overline{B_0}$

So the minimal expressions to be implemented using PLA are

$E_3 = B_3 + B_2B_0 + B_2B_1$; $E_2 = B_2\overline{B_1}\,\overline{B_0} + \overline{B_2}B_0 + \overline{B_2}B_1$; $E_1 = \overline{B_1}\ \overline{B_0} + B_1B_0$; $E_0 = \overline{B_0}$

The realization using PLA is shown in figure below.

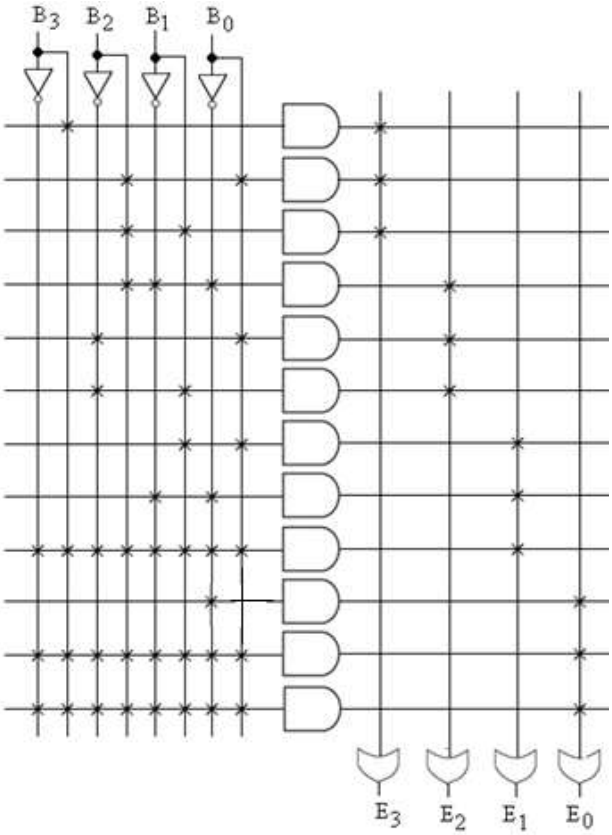


Implementation using PLA

c) The minimal expressions for E_3, E_2, E_1 and E_0 to be implemented using PAL are

$E_3 = B_3 + B_2B_0 + B_2B_1$; $E_2 = B_2\overline{B_1}\,\overline{B_0} + \overline{B_2}B_0 + \overline{B_2}B_1$; $E_1 = \overline{B_1}\ \overline{B_0} + B_1B_0$; $E_0 = \overline{B_0}$

The realization of BCD to Excess-3 code converter using PAL is shown in figure below.



Realization of BCD to Excess-3 Code converter using PAL

2 Design an XS-3-to-BCD code converter using a (a) PROM, (b) PLA and (c) PAL

Solution:

a) The conversion table for Excess-3 to BCD code converter is shown below.

Conversion Table

Excess-3				BCD			
E_3	E_2	E_1	E_0	B_3	B_2	B_1	B_0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1

Since terms 0000, 0001, 0010, 1101, 1110, 1111, i.e. minterms 0, 1, 2, 13, 14, 15 are don't cares in Excess-3, from the truth table we get

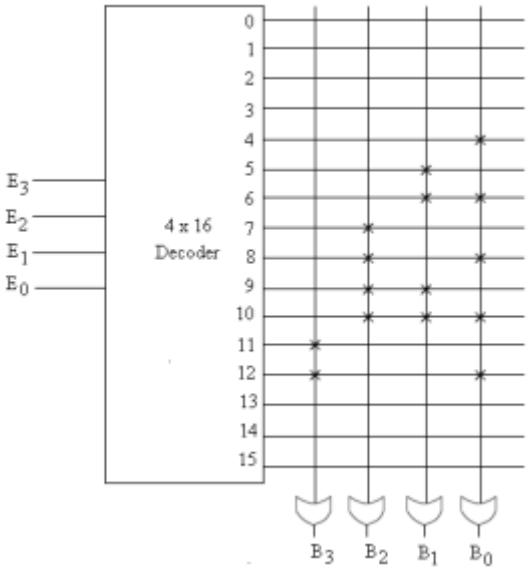
$$B_3 = \sum m\,(11, 12) + d(0, 1, 2, 13, 14, 15)$$

$$B_2 = \sum m\,(7, 8, 9, 10) + d(0, 1, 2, 13, 14, 15)$$

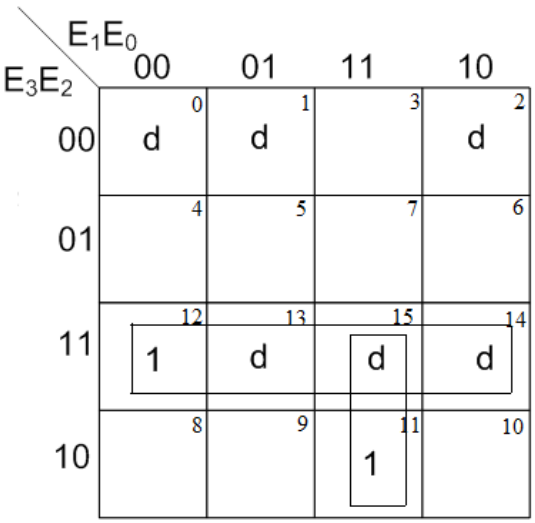
$$B_1 = \sum m\,(5, 6, 9, 10) + d(0, 1, 2, 13, 14, 15)$$

$$B_0 = \sum m\ (4, 6, 8, 10, 12) + d(0, 1, 2, 13, 14, 15)$$

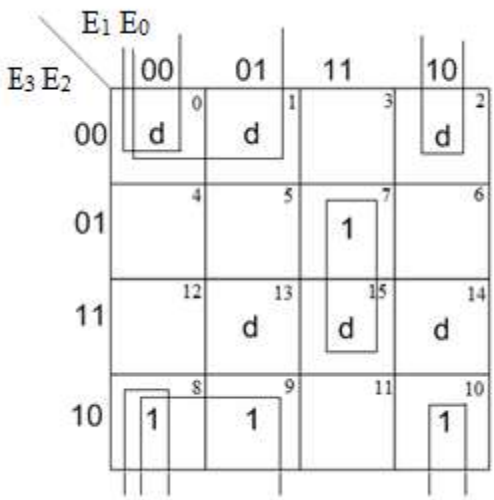
For realization by EPROM, there is no need to minimize the functions B_3, B_2, B_1, and B_0. Since the input is a 4-variable one, we require a 4×16 decoder. Since the number of outputs is 4, we require a 16×4 PROM. The realization of the Excess-3-to-BCD code converter using a PROM is shown in figure below.



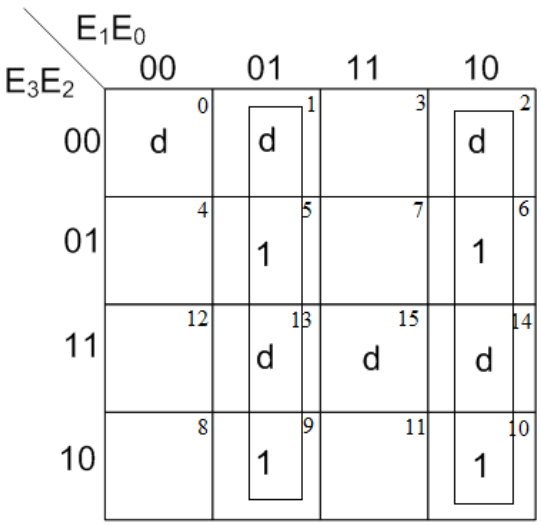
b) The minimal expressions for B_3, B_2, B_1 and B_0 are obtained using K-maps as shown below.



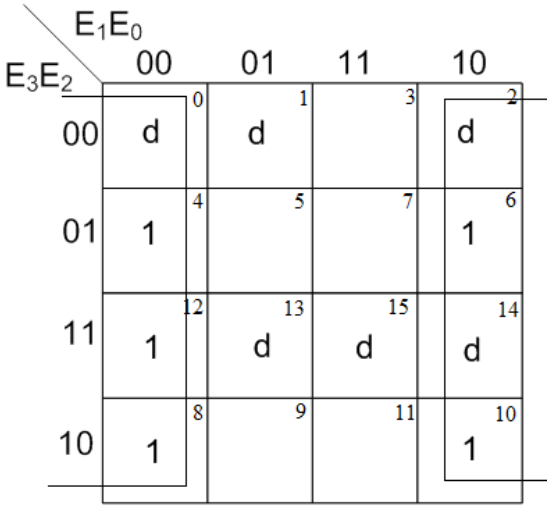
$$B_3 = E_3E_2 + E_3E_1E_0$$



$$B_2 = \overline{E_2}\ \overline{E_0} + \overline{E_2}\ \overline{E_1} + E_2E_1E_0$$



$$B_1 = \overline{E_1}E_0 + E_1\overline{E_0}$$

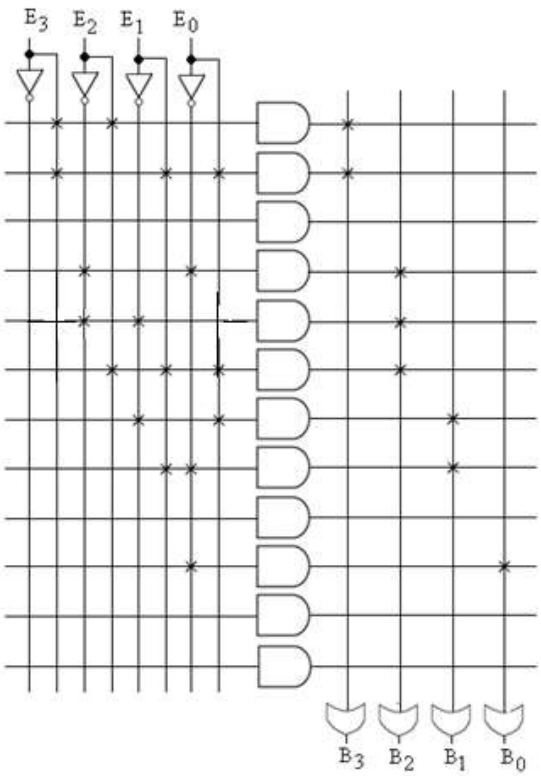


$$B_0 = \overline{E_0}$$

So the minimal expressions to be implemented using PLA are

$B_3 = E_3E_2 + E_3E_1E_0$; $B_2 = \overline{E_2}\ \overline{E_0} + \overline{E_2}\ \overline{E_1} + E_2E_1E_0$; $B_1 = \overline{E_1}\ E_0 + E_1\overline{E_0}$; $B_0 = \overline{E_0}$

The realization of Excess-3 to BCD code converter using PLA is shown in figure below.

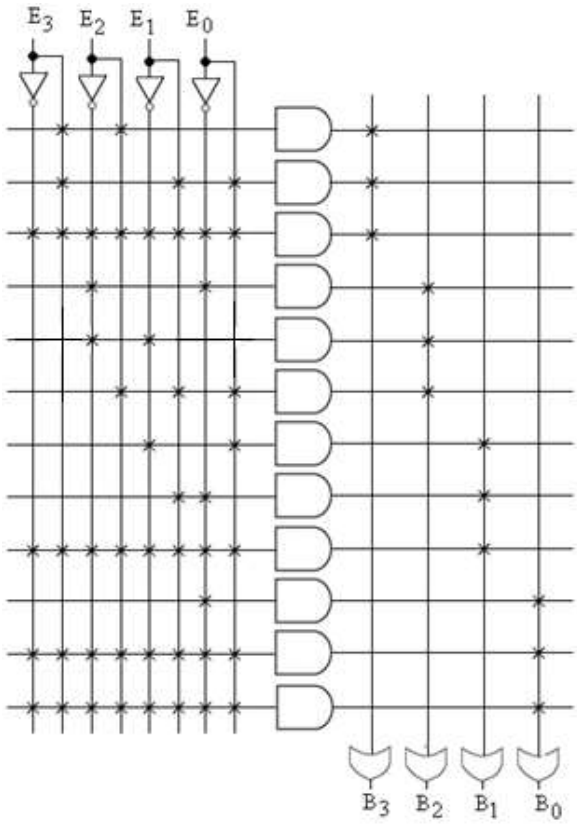


Implementation of an Excess-3 to BCD code converter using PLA

c) The minimal expressions for B_3, B_2, B_1 and B_0 are as shown below.

$B_3 = E_3E_2 + E_3E_1E_0$; $B_2 = \overline{E_2}\ \overline{E_0} + \overline{E_2}\ \overline{E_1} + E_2E_1E_0$; $B_1 = \overline{E_1}\,E_0 + E_1\overline{E_0}$; $B_0 = \overline{E_0}$

The realization of Excess-3 to BCD code converter using PAL is shown in figure below.



Realization of Excess-3 to BCD code converter using PAL

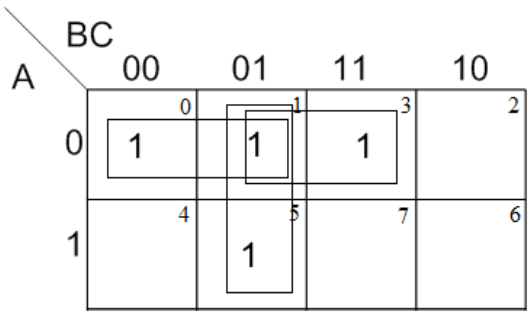
3 Implement the following Boolean functions using PLA

$$f_1(A,B,C) = \sum m(0,1,3,5)\,;\; f_2(A,B,C) = \sum m(0,3,5,7)$$

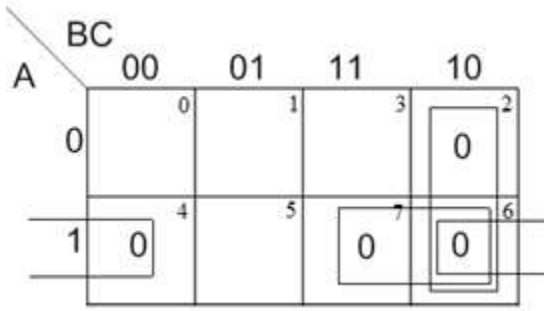
Solution:

Given $f_1(A,B,C) = \sum m(0,1,3,5)\,;\; f_2(A,B,C) = \sum m(0,3,5,7)$

The K-maps for the functions f_1 and f_2, their minimization and the minimal expressions for both the true and complement forms of those in SOP form are shown in figure below.

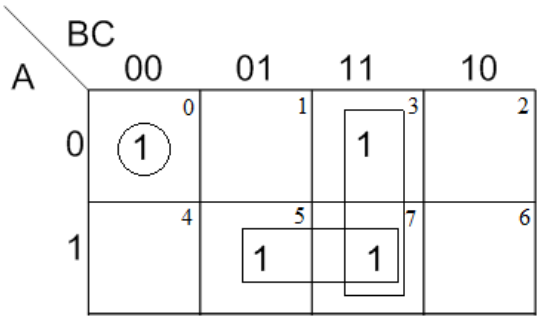


$$f_1(T) = \bar{A}\,\bar{B} + \bar{A}\,C + \bar{B}C$$

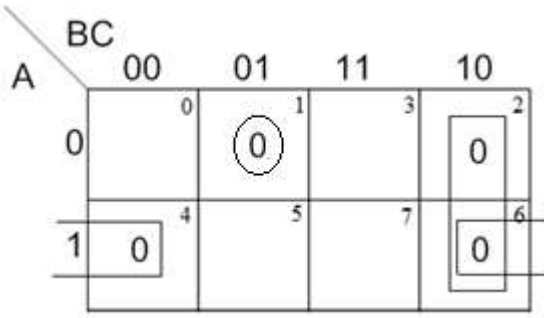


$$\overline{f_1} = A\bar{C} + AB + B\bar{C}$$

$$f_1(C) = \overline{A\bar{C} + AB + B\bar{C}}$$



$$f_2(T) = \bar{A}\,\bar{B}\,\bar{C} + AC + BC$$



$$\overline{f_2} = \bar{A}\,\bar{B}\,C + A\bar{C} + B\bar{C}$$

$$f_2(C) = \overline{\bar{A}\,\bar{B}\,C + A\bar{C} + B\bar{C}}$$

Out of $f_1(T), f_1(C), f_2(T)$ and $f_2(C)$, the combination that gives minimum number of product terms is

$$f_1(C) = \overline{A\bar{C} + AB + B\bar{C}} \text{ and } f_2(C) = \overline{\bar{A}\,\bar{B}\,C + A\bar{C} + B\bar{C}}$$

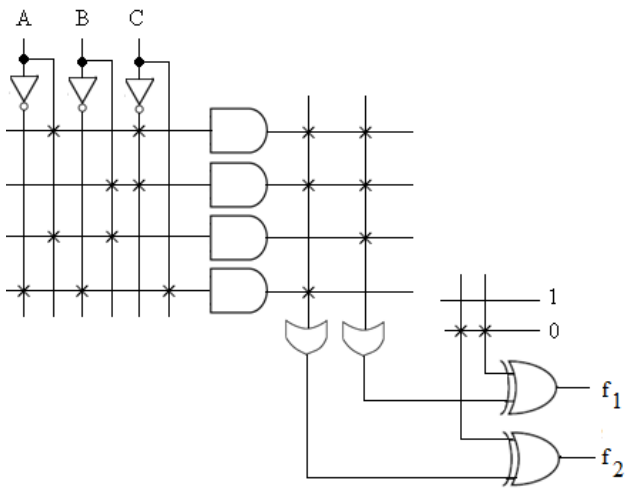
This gives 4 distinct terms: $A\bar{C}, B\bar{C}, AB, \bar{A}\,\bar{B}C$.

The PLA programming table for this combination is shown in figure.

Product Term		Inputs			Outputs (C)	Outputs (C)
		A	B	C	f_1	f_2
1	$A\bar{C}$	1	-	0	1	1
2	$B\bar{C}$	-	1	0	1	1
3	AB	1	1	-	1	-
4	$\bar{A}\,\bar{B}\,C$	0	0	1	-	1

(a) PLA Programming Table

The implementation using a PLA is shown in figure below.



(b) Fuse

Map

4 Implement the following Boolean functions using PAL.

$$f_1(w,x,y,z) = \sum m(0,2,5,7,8,10,12,13)$$

$$f_2(w,x,y,z) = \sum m(0,2,6,8,9,14,15)$$

$$f_3(w,x,y,z) = \sum m(0,8,14,15)$$

$$f_4(w,x,y,z) = \sum m(0,1,2,4,5,8,9,10)$$

Solution:

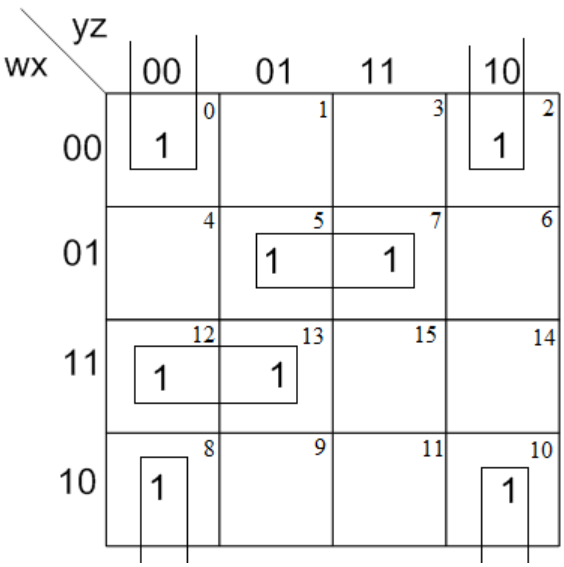
Given $f_1(w,x,y,z) = \sum m(0,2,5,7,8,10,12,13)$

$$f_2(w,x,y,z) = \sum m(0,2,6,8,9,14,15)$$

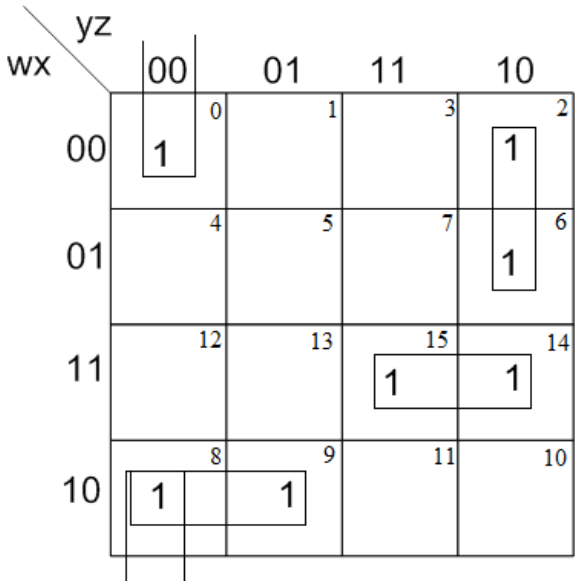
$$f_3(w,x,y,z) = \sum m(0,8,14,15)$$

$$f_4(w,x,y,z) = \sum m(0,1,2,4,5,8,9,10)$$

The minimal expressions for f_1, f_2, f_3 and f_4 are obtained using K-maps as shown below.

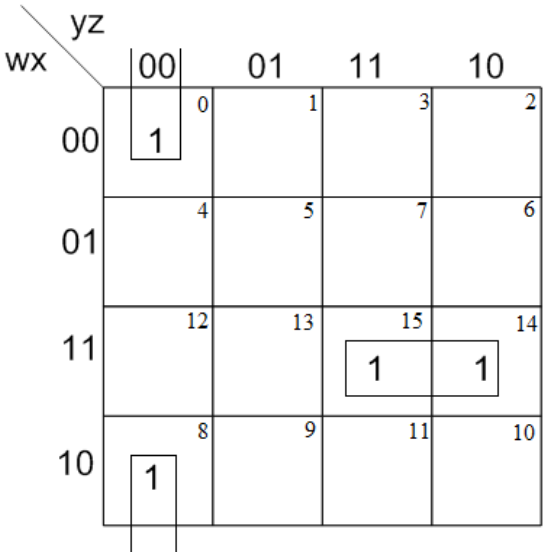


$f_1 = \bar{x}\,\bar{z} + wx\bar{y} + \bar{w}xz$

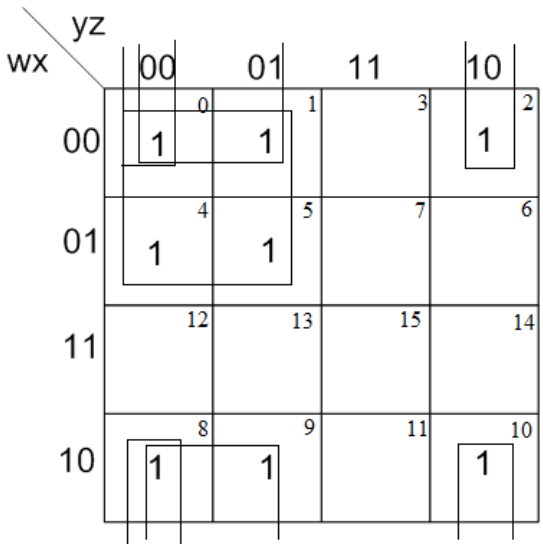


$f_2 = \bar{x}\,\bar{y}\,\bar{z} + w\bar{x}\,\bar{y} + wxy + \bar{w}y\bar{z}$

$= f_3 + w\bar{x}\,\bar{y} + \bar{w}y\bar{z}$



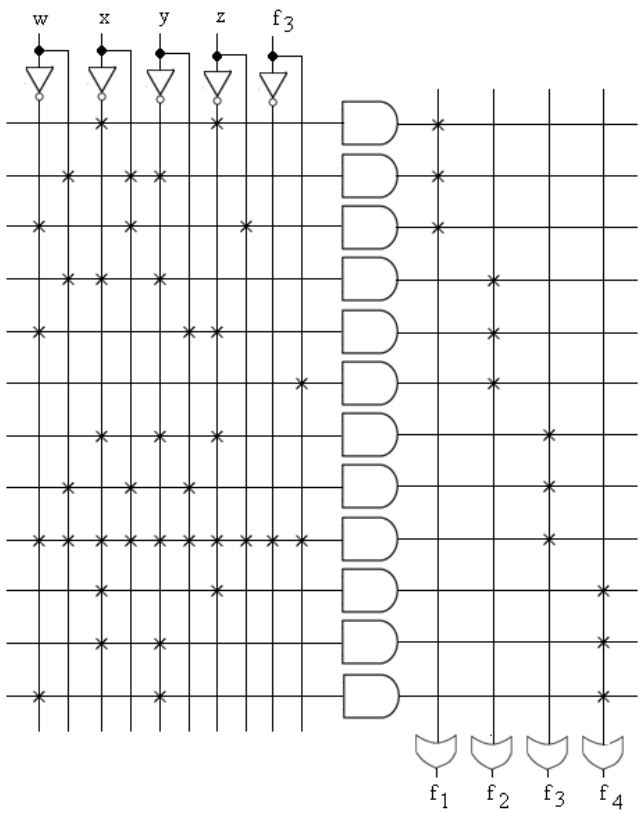
$f_3 = \bar{x}\,\bar{y}\,\bar{z} + wxy$



$f_4 = \bar{x}\,\bar{z} + \bar{x}\,\bar{y} + \bar{w}\bar{y}$

The PAL programming table and the logic diagram to realize f_1, f_2, f_3 and f_4 are shown in figure below.

Product Term	AND Inputs					Outputs
	w	x	y	z	f_3	
1	-	0	-	0	-	
2	1	1	0	-	-	$f_1 = \bar{x}\,\bar{z} + wx\bar{y} + \bar{w}xz$
3	0	1	-	1	-	
4	1	0	0	-	-	
5	0	-	1	0	-	$f_2 = w\bar{x}\,\bar{y} + \bar{w}y\,\bar{z} + f_3$
6	-	-	-	-	1	
7	-	0	0	0	-	
8	1	1	1	-	-	$f_3 = \bar{x}\,\bar{y}\,\bar{z} + wxy$
9	-	-	-	-	-	
10	-	0	-	0	-	
11	-	0	0	-	-	$f_4 = \bar{x}\,\bar{y} + \bar{x}\,\bar{z} + \bar{w}\bar{y}$
12	0	-	0	-	-	



Realization using PAL

5 Tabulate the PLA programming table for the four Boolean functions listed below.

$f_1(A,B,C) = \sum m(0,1,2,4,6)$; $f_2(A,B,C) = \sum m(0,2,6,7)$

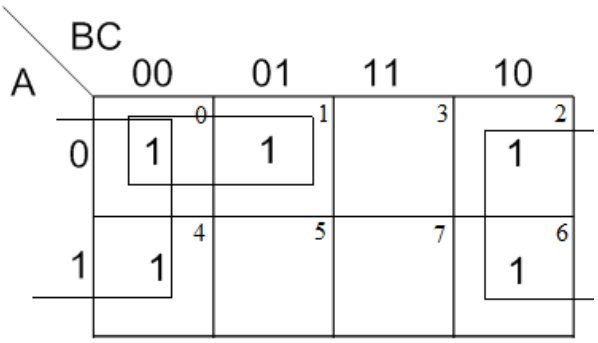
$f_3(A,B,C) = \sum m(3,6)$; $f_4(A,B,C) = \sum m(1,3,5,7)$

Solution:

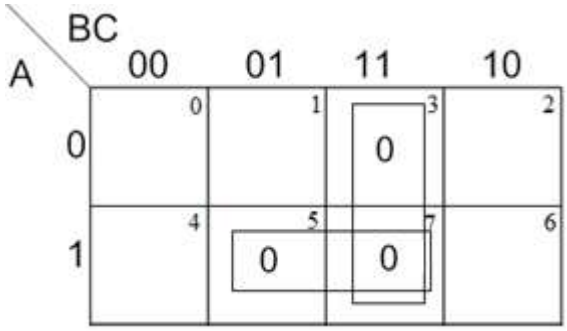
Given $f_1(A,B,C) = \sum m(0,1,2,4,6)$; $f_2(A,B,C) = \sum m(0,2,6,7)$;

$f_3(A,B,C) = \sum m(3,6)$; $f_4(A,B,C) = \sum m(1,3,5,7)$

The K-maps for f_1, f_2, f_3, f_4, their minimization and the minimal expressions obtained in true and complement form are shown in figure below.

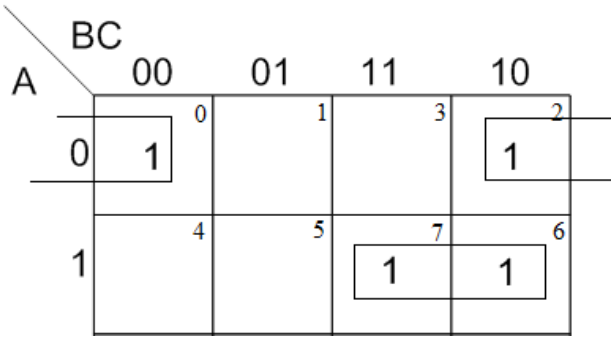


$f_1(T) = \bar{C} + \bar{A}\,\bar{B}$

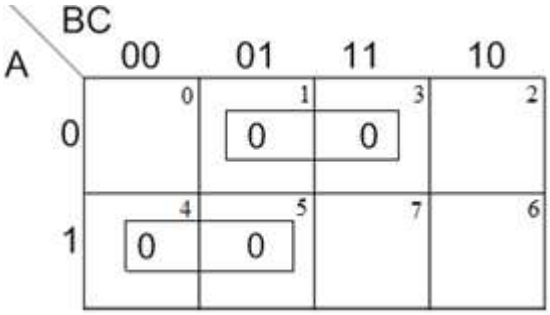


$\bar{f_1} = AC + BC$

$f_1(C) = \overline{AC + BC}$

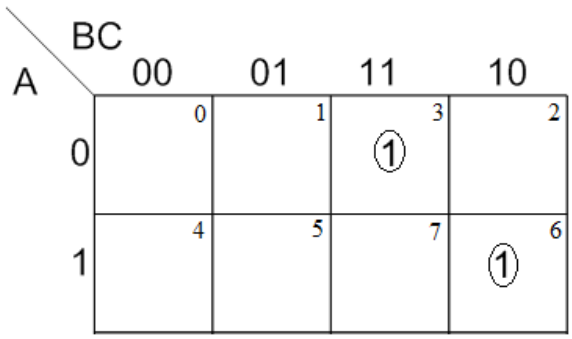


$f_2(T) = \bar{A}\,\bar{C} + AB$

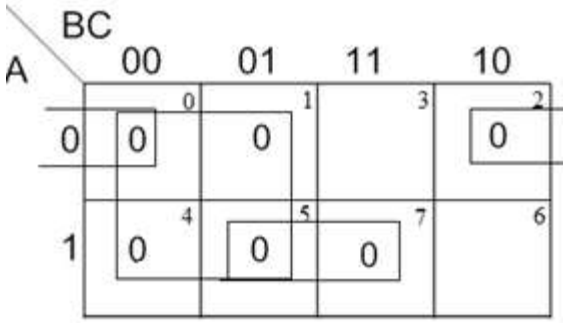


$\bar{f_2} = A\bar{B} + \bar{A}\,C$

$f_2(C) = \overline{A\bar{B} + \bar{A}\,C}$

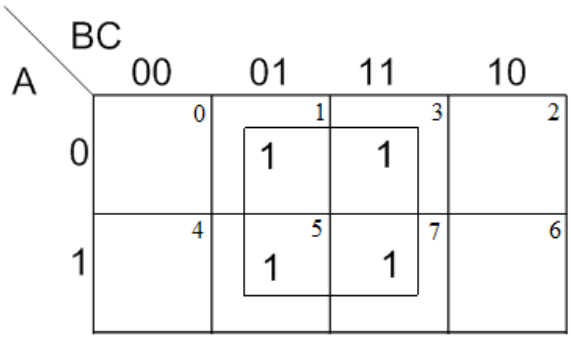


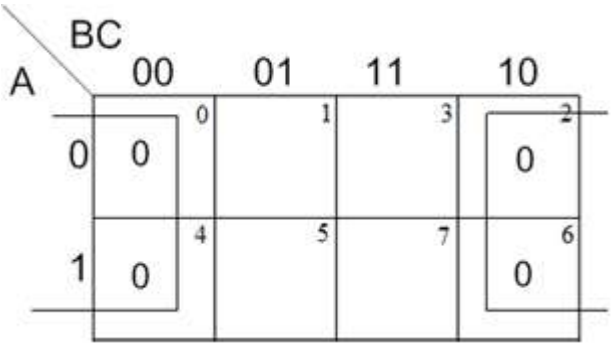
$f_3(T) = \bar{A}\,BC + AB\bar{C}$



$\bar{f_3} = \bar{B} + AC + \bar{A}\,\bar{C}$

$f_3(C) = \overline{\bar{B} + AC + \bar{A}\,\bar{C}}$





$f_4(T) = C$ $\qquad\qquad$ $\bar{f}_4 = \bar{C}; f_4(C) = \bar{\bar{C}} = C$

Out of $f_1(T), f_1(C), f_2(T), f_2(C), f_3(T), f_3(C), f_4(T)$ and $f_4(C)$, the combination that gives minimum number of product terms is

$f_1(C) = \overline{AC + BC}$; $f_2(T) = \bar{A}\,\bar{C} + AB; f_3(C) = \overline{\bar{B} + AC + \bar{A}\,\bar{C}}$ and $f_4(T) = C$.

This gives 6 distinct terms: $AC, BC, \bar{A}\,\bar{C}, AB, \bar{B}$, and C. The PLA programming table for this combination is shown in figure below.

Product Term		Inputs			Outputs (C)	(T)	(C)	(T)
		A	B	C	f_1	f_2	f_3	f_4
1	AC	1	-	1	1	-	1	-
2	BC	-	1	1	1	-	-	-
3	$\bar{A}\,\bar{C}$	0	-	0	-	1	1	-
4	AB	1	1	-	-	1	-	-
5	$\bar{B}$	-	0	-	-	-	1	-
6	C	-	-	1	-	-	-	1